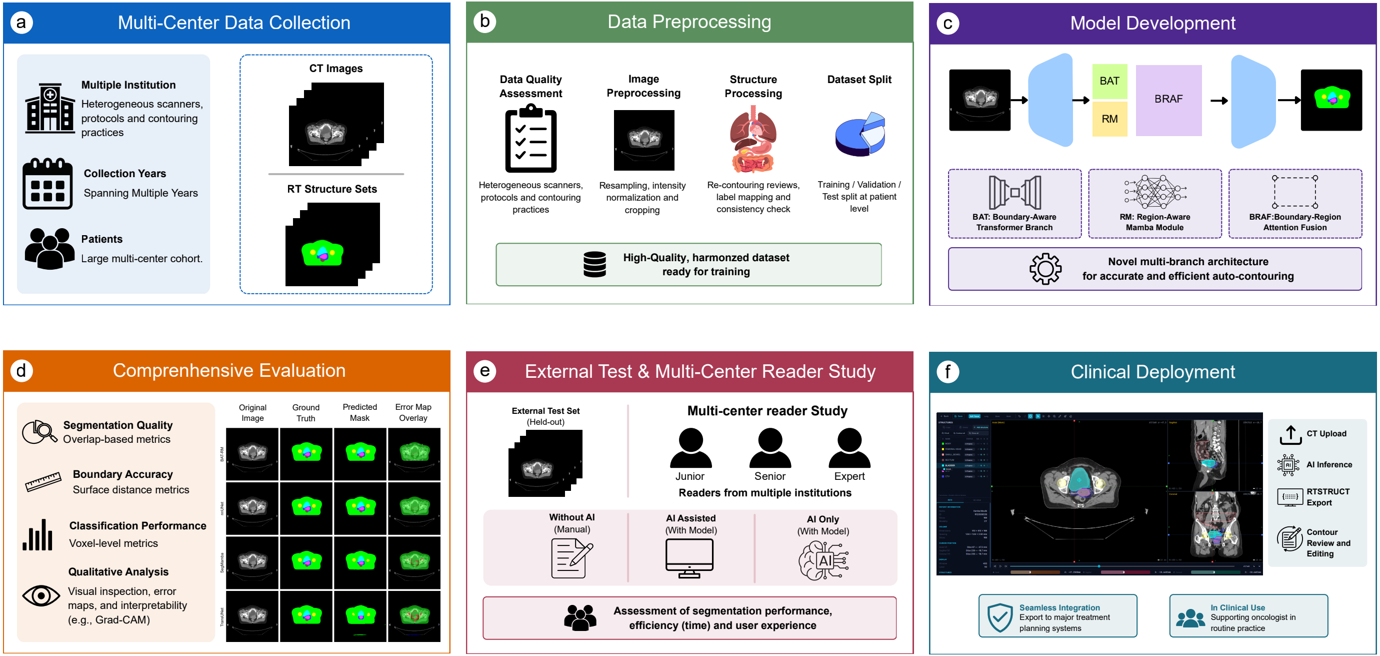

a
Multi-Center Data Collection
Multiple Institution
Heterogeneous scanners, protocols and contouring practices
Collection Years
Spanning Multiple Years
Patients
Large multi-center cohort.
CT Images
RT Structure Sets
b
Data Preprocessing
Data Quality Assessment
Heterogeneous scanners, protocols and contouring practices
Image Preprocessing
Resampling, intensity normalization and cropping
Structure Processing
Re-contouring reviews, label mapping and consistency check
Dataset Split
Training / Validation / Test split at patient level
High-Quality, harmonzed dataset ready for training
c
Model Development
BAT
RM
BRAF
BAT: Boundary-Aware Transformer Branch
RM: Region-Aware Mamba Module
BRAF:Boundary-Region Attention Fusion
Novel multi-branch architecture for accurate and efficient auto-contouring
d
Comprenhensive Evaluation
Segmentation Quality
Overlap-based metrics
Boundary Accuracy
Surface distance metrics
Classification Performance
Voxel-level metrics
Qualitative Analysis
Visual inspection, error maps, and interpretability (e.g., Grad-CAM)
Original Image
Ground Truth
Predicted Mask
Error Map Overlay
e
External Test & Multi-Center Reader Study
External Test Set (Held-out)
Multi-center reader Study
Junior
Senior
Expert
Readers from multiple institutions
Without AI (Manual)
AI Assisted (With Model)
AI Only (With Model)
AI
Assessment of segmentation performance, efficiency (time) and user experience
f
Clinical Deployment
CT Upload
AI Inference
RTSTRUCT Export
Contour Review and Editing
Seamless Integration
Export to major treatment planning systems
In Clinical Use
Supporting oncologist in routine practice